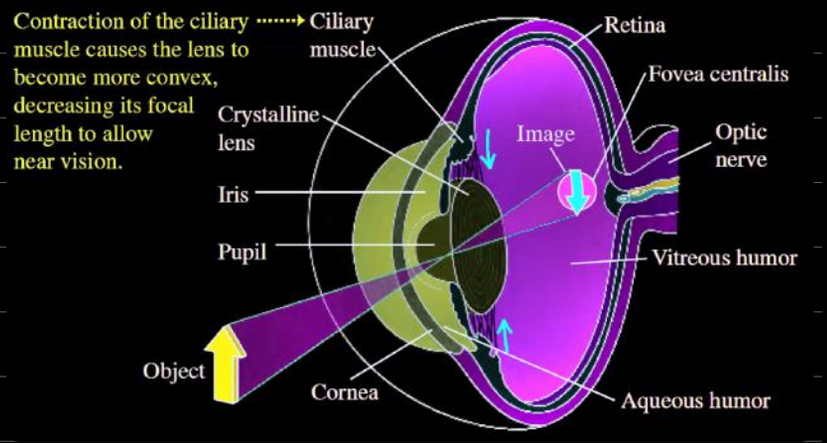
## HUMAN EYE:

A schematic diagram of human eye is shown in the adjacent figure. Here the Retina is about 2.5 cm behind the eye lens. Since the "screen distance" is fixed, to focus different objects on Retina, we have a lens of variable focal length, which is controlled by ciliary muscles. To look at far away objects, a small converging power is required because of smaller divergence angle in the pencil of rays entering the eye, whereas for nearby points, the pencil of rays is highly divergent and we need greater converging power to converge the rays at the same position.




The pupil controls the amount of light that enters the eyes to adjust for different lighting conditions.

**NEAR POINT:** To clearly see the objects we need to squeeze our eye lens (to bulge it or reduce its radius of curvature) as the objects are brought nearer and nearer. However beyond a certain point, we can't focus on the object properly even with fully strained ciliary muscles. This point is called near point of the eye. For a normal human eye this distance is taken as $D = 25$ cm.

**FAR POINT:** The farthest our eyes can see clearly by fully relaxing the ciliary muscles is called the far point of the eye. For a normal human eye, the far point is infinity.

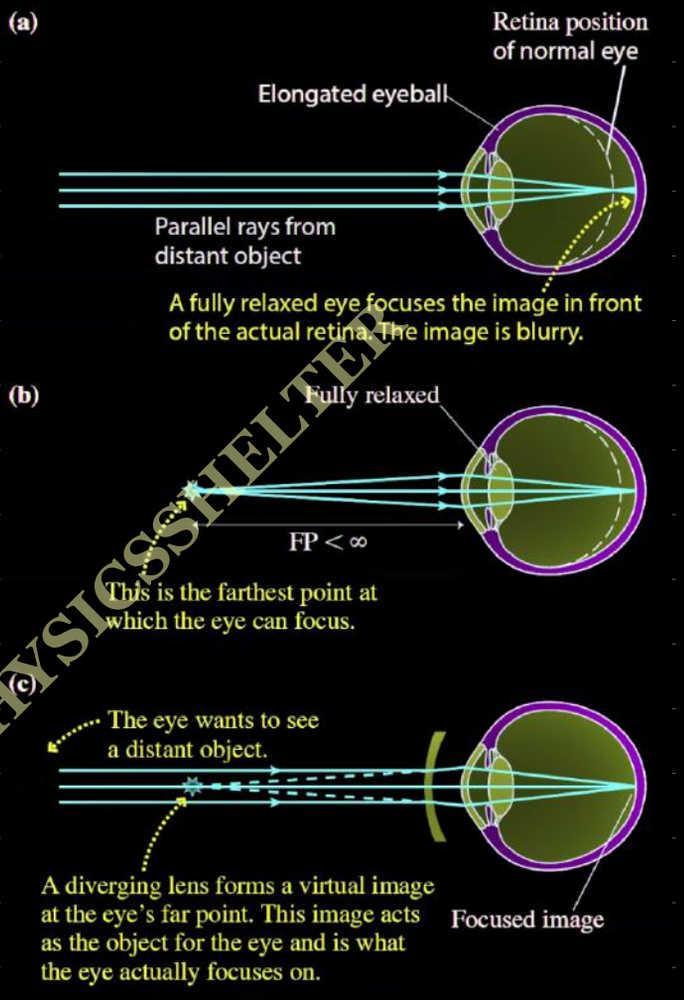
**ACCOMODATION:** The process of adjusting the focal length of the eye lens as the object is moved from near point to the far point is called accomodation.

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# Eye Defects And Their Corrections:

## Myopia (Near Sightedness): [Near Point ok Far Point < ∞):

As discussed earlier, we need to have smaller converging power to focus on distant objects and a larger power to focus on nearby objects. For a myopic person, the power of a fully relaxed eye is more than that required for focusing at infinity. Thus, there is a limit to his far point. For correcting this defect we use diverging lens which forms the Image of ∞ at person's far point and then the person is able to process it with his eyes. Due to more than required converging power, the unaided myopic eye converges the rays in front of the retina. The diverging lens reduces the effective converging power thus helping the person form the image at the retina.




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**Example:** If your far point is 200 cm, what power of spectacles should you use for correcting the vision defect.

**Solution:** Here we must form the image of infinity at 200 cm. Thus using lens formula:

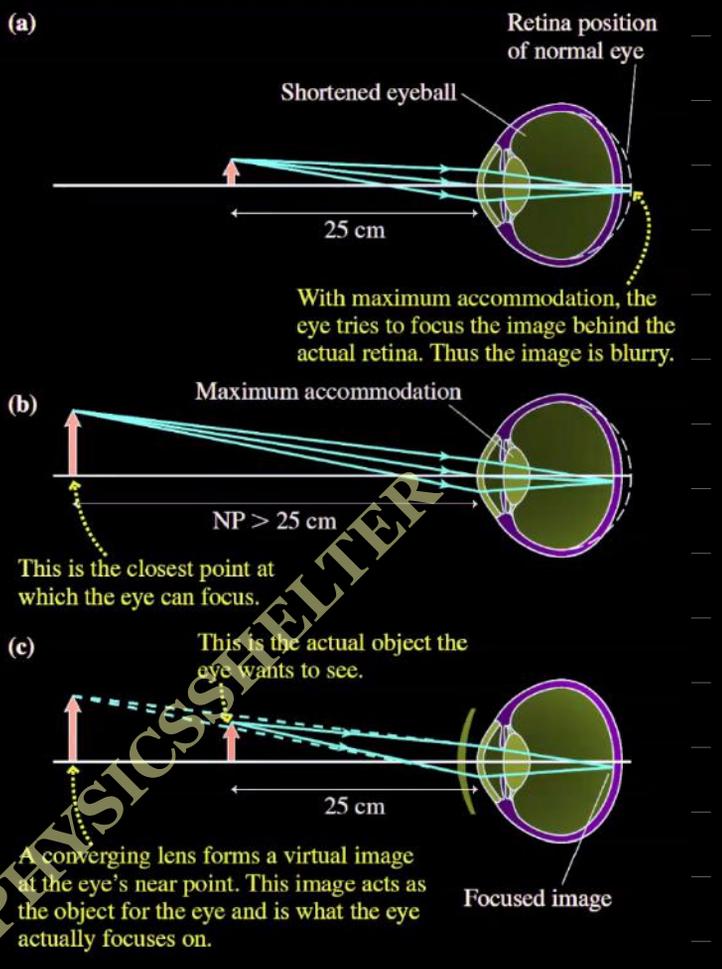
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-2} - \frac{1}{\infty} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} = -0.5 \quad \text{thus required power of lens is } -0.5 \text{ Diopters.}$$

→ Near point actually keeps receding with age. On an average a 10 year old has a near point of 7cm and a 60 year old has 200 cm.

## HYPEROPIA (HYPERMETROPIA): FAR POINT OK NEAR POINT > 25 cm:

As discussed earlier, we need a higher converging power to focus on a nearby object than that for a far away object. A hyperopic person has less than the required convergence power to see an object 25 cm away from the eye. This person's near point is more than 25 cm. Thus to correct this vision defect we use converging lens. The converging lens forms the image of a 25 cm distant object at person's near point and thus enabling him to see the object.




**EXAMPLE:** Suppose the near point of your left eye is 150 cm. What lens prescription will restore the normal vision?

**Solution:** Here we need to form image of 25 cm distant object at 150 cm. Thus:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-1.5} - \frac{1}{-0.25} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{10}{3}\ m^{-1}$$

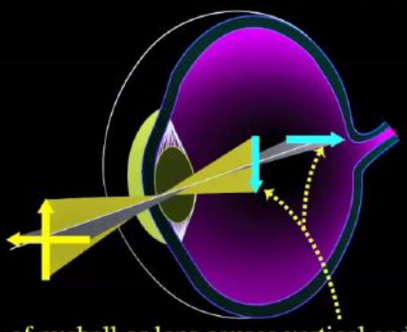
Thus required lens power is +3.33 D (dioptres).

**PRESBYOPIA:** When both near point and far point are defective, the condition is known as presbyopia. To correct this vision defect we use either bifocal lenses (Glass piece with two distinct focal lengths in upper and lower parts) or we use progressive lenses (continuously varying focal lengths).

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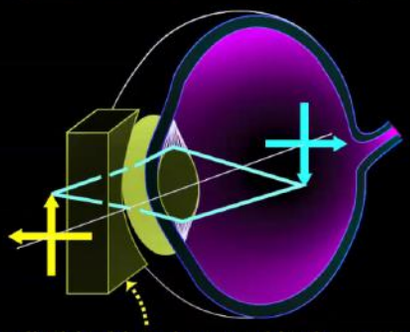
## Astigmatism:

(a) Vertical lines are imaged in front of the retina.



Shape of eyeball or lens causes vertical and horizontal elements to focus at different distances.

(b) A cylindrical lens corrects for astigmatism.



This cylindrical lens is curved in the vertical, but not the horizontal, direction; it changes the focal length of vertical elements.

This defect arises because the eye lens is more sharply curved in one plane than the other, thus the horizontal and vertical elements are focused at different places. To correct this defect, we use cylindrical lenses. For instance in the above figure, the concave cylindrical lens increases the effective focal length for vertical elements.

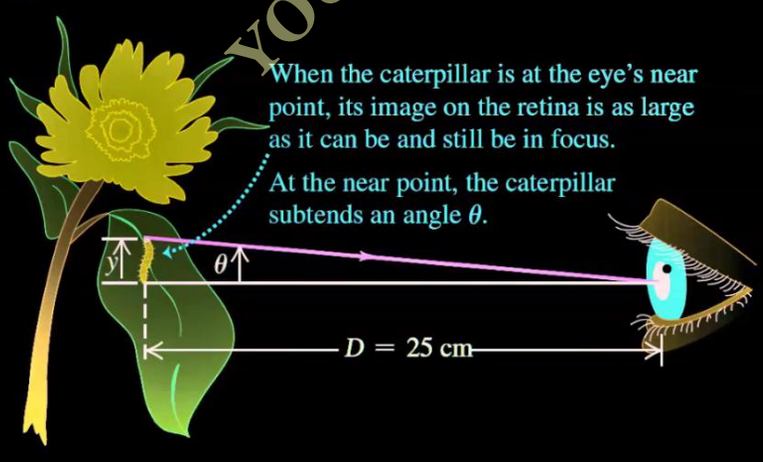
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## Simple Magnifier (Simple Microscope):

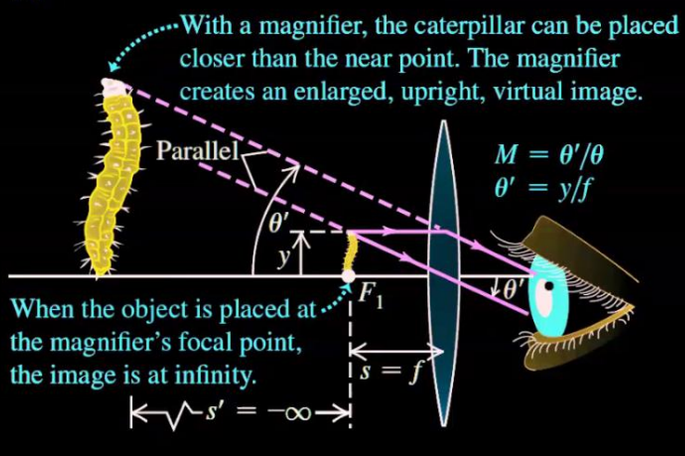
In case of optical instruments, by magnification we usually mean angular magnification, i.e ratio of angle subtended by the image of an object with help of that device to the maximum angle subtended on the unaided eye (i.e at near point for microscopes).

## Simple Microscope In Normal Adjustment:

(a)




(b)



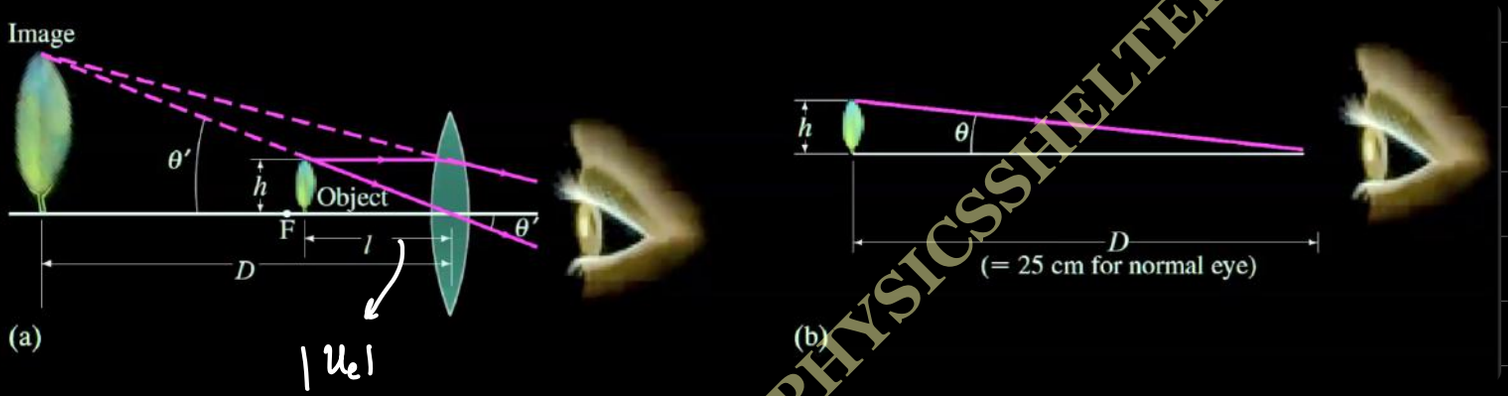

When an object is positioned so as to see the image with a relaxed eye, we call it a normal adjustment (i.e image

is formed at ∞). Suppose we wish to see a caterpillar of length y closely, then maximum angle it can subtend on our eye, $\theta = y/D$. Now if we see it through the magnifier in normal adjustment, we can place it at the focus of our lens (second figure) and then angle subtended on our eye becomes $\theta' = y/f$. Now angular magnification:

$M = \theta'/\theta = D/f$ thus for normal adjustment:

$$\boxed{M = D/f}$$

## Simple Microscope In Near Point Adjustment:




When an object is placed so as to see the image with most strained eye, we call it the near point adjustment. Let h be the height of an object then for an uncided eye (figure b), $\theta = h/D$ while for the aided eye, $\theta' = h/|u_e|$ where $|u_e|$ is the object location for near point adjustment (figure a). Thus angular magnification:

$M = \theta'/\theta = D/|u_e|$, now let us find $|u_e|$. Using lens formula,

$$\frac{1}{-D} - \frac{1}{u_e} = \frac{1}{f} \Rightarrow |u_e| = \frac{Df}{D+f}, \text{ thus,}$$

$M = D/|u_e| = 1 + D/f$ thus for near point adjustment:

$$\boxed{M = 1 + D/f = D/|u_e|}$$

**Example:** A watchmaker's lens has 8cm focal length. Find its magnification in (a) Normal adjustment (b) Near point adjustment.

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**Solution:** As discussed in previous sections, for normal adjustment,

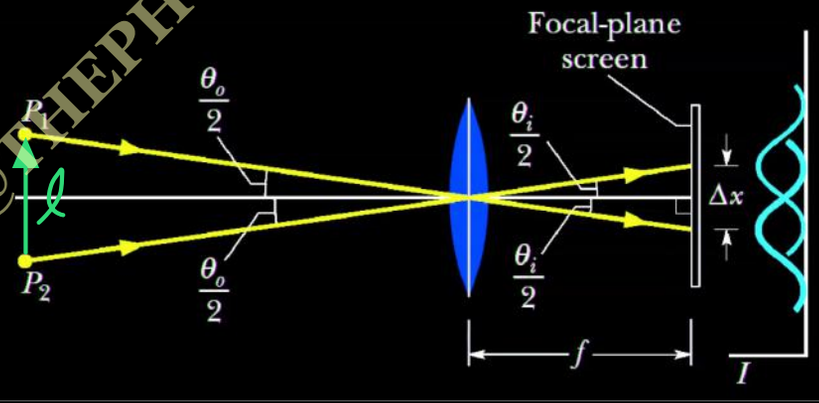
$M = D/f = 25/8 = 3.125$ Ans.

For near point adjustment, $M = 1 + D/f = 4.125$ Ans.

→ Note that we can have infinitely many adjustments between near point and normal adjustments and we can find their magnifications using the same method.

## Resolving Power Of A Simple Microscope:

Recall from the chapter on wave optics that whenever we form images of point objects through a lens or a circular aperture, it has an angular spread of $1.22\lambda/d$ due to diffraction effects. Considering Fraunhofer diffraction, using Rayleigh's criterion, we found that minimum angular separation between the objects for getting clearly resolved is $\theta_{min} = 1.22\lambda/d$ where $d$ is the lens aperture. The minimum required separation between the objects can be found by multiplying the object distance by minimum required angular separation. Thus for normal adjustment, $\ell_{min} = \frac{1.22\lambda}{d} \times f$. The



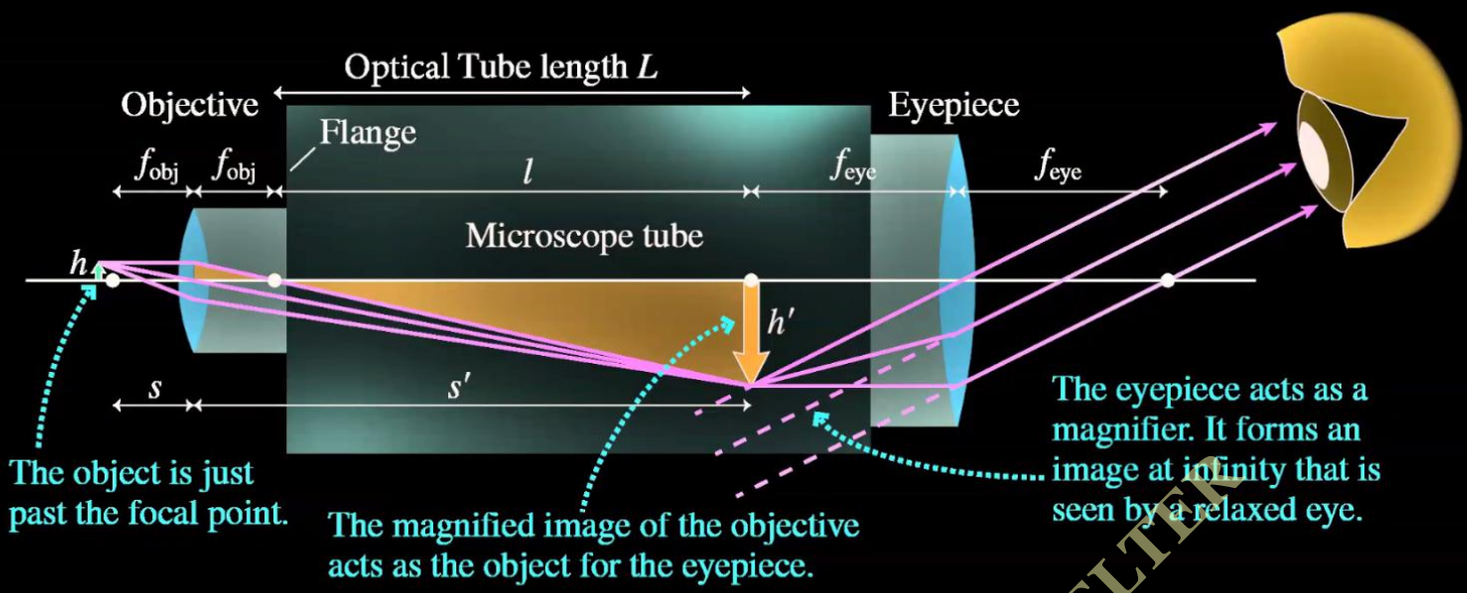

smaller the value of $\ell_{min}$, we say higher the resolving power, thus sometimes resolving power is reported in $m^{-1}$ as $R.P. = \frac{1}{\ell_{min}}$.

(For other adjustments, resolving power can be found similarly).

→ Student is advised to review fraunhofer diffraction from circular apertures and Rayleigh's criterion.

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## Compound Microscope : (Normal Adjustment):




The figure above shows schematic of a compound microscope under normal adjustment. Here the objective lens produces linear magnification and then the eyepiece behaves just like a normal magnifier. Suppose $h$ is the height of an object then maximum angle that can be subtended on a normal eye is $\theta = h/D$. Now with compound microscope, intermediate image is formed at the focus of the eyepiece, thus $\theta' = h'/f_e$ where $f_e$ is the focal length of the eyepiece. Also magnification of objective using Newton's convention is $m = L/f_o$ where $L$ is the optical tube length [ Distance between foci of the two lenses). Thus $h' = -hL/f_o$ thus, $\theta' = -hL/f_o f_e$ Thus angular magnification from compound microscope:

$$M = \frac{\theta'}{\theta} = \frac{-hL}{f_o f_e} \Big/ \left(\frac{h}{D}\right) = \frac{-L}{f_o} \times \frac{D}{f_e} = m_o M_e$$

i.e product of linear magnification of objective and angular magnification of the eyepiece. Thus:

$$\boxed{M = \frac{-L}{f_o} \times \frac{D}{f_e}}$$ ← Inverted Image ∴ -Ve Sign.

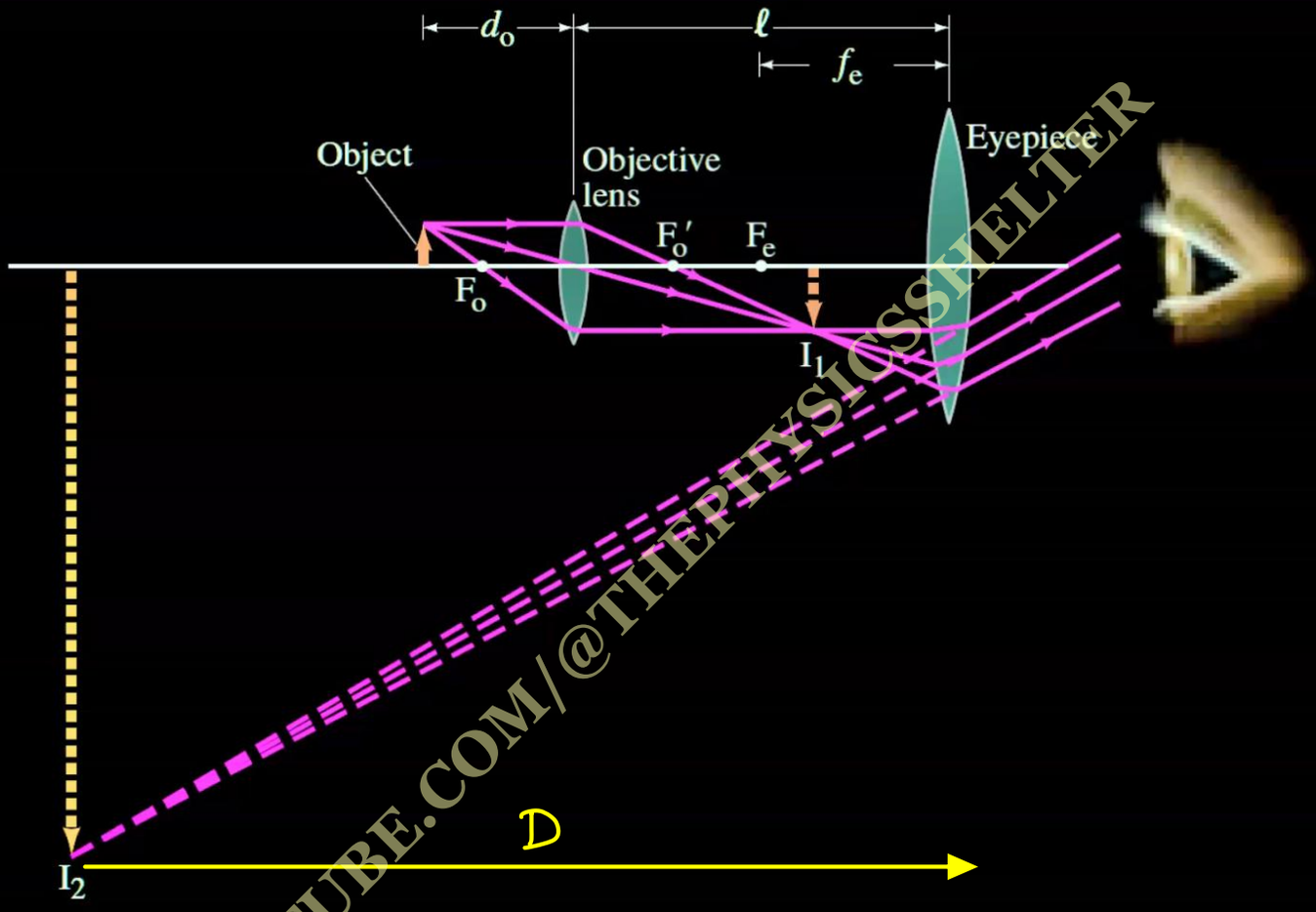
→ In a typical microscope, focal length of the objective is small and object is kept very near to the focus so as to obtain

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maximum possible linear magnification through it. Thus if data appears insufficient, use $|u_o| \approx f_o$.

→ NCERT Refers to optical tube length as just tube length, whereas in JEE main questions, tube length ambiguously means either mechanical tube length or optical tube length (though mostly optical tube length)

## Compound Microscope (Near Point Adjustment):




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For near point adjustment also we can find the angular magnification as $M = m_o M_e$ where $m_o$ is linear magnification of objective and $M_e$ is angular magnification of the eyepiece, thus the magnification in near point adjustment can be written as:

$$M = \frac{v_o}{u_o} \times \left(1 + \frac{D}{f_e}\right) = \frac{v_o}{u_o} \times \frac{D}{|u_e|}$$

(Recall that angular magnification of simple magnifier in near point adjustment was $1 + D/f$).

→ For any arbitrary adjustment also we can find the magnification as $M = m_o m_e$, just keep the basic ray diagram in mind.

**EXAMPLE : (JEE Modified):** If we need magnification of 375 from a microscope of mechanical tube length 150 mm and objective of focal length 5 mm, what should be the focal length of eye piece? Assume Normal adjustment.

**Solution:** $M = \frac{L}{f_o} \times \frac{D}{f_e}$ where $L$ is optical tube length.

Now, $f_o + L + f_e = 150$ ① (mechanical tube length)

$\Rightarrow$ $5\text{mm} + L + f_e = 150\text{ mm}$

and $\frac{L}{5} \times \frac{250}{f_e} = 375$ ② (everything in mm)

Solving ① & ②: $L \simeq 129$ mm, $f_e \simeq 17$ mm

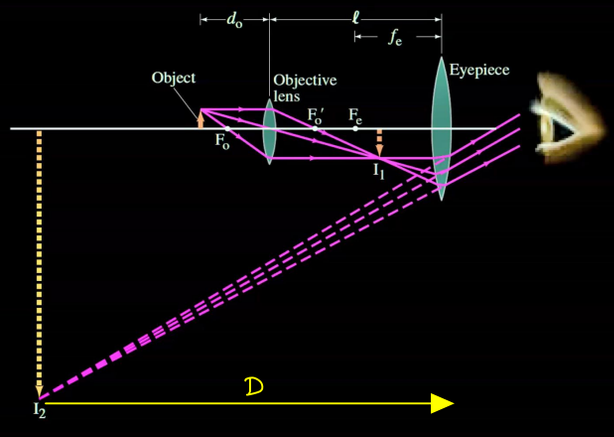
**EXAMPLE :** Repeat the previous problem if the final image is to be formed at least distance of distinct vision.

**Solution:** Let intermediate image be formed at $x$ distance from focus of objective then:

$f_o + x + |u_e| = 150\text{ mm}$

i.e: $x + \frac{Df_e}{D+f_e} = 145\text{ mm}$ ①

and $\frac{x}{f_o} \times \left(1 + \frac{D}{f_e}\right) = 375$ ②




① $\Rightarrow$ $x + \frac{250 f_e}{250 + f_e} = 145\text{ mm}$

② $\Rightarrow$ $\frac{x}{5} \times \frac{250 + f_e}{f_e} = 375$

This gives $x = 127$ mm, $f_e = 18$ mm also

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**EXAMPLE :** Repeat the previous problem if instead of mechanical tube length, optical tube length were 150 mm and final image was formed at ∞.

**Solution:** Now $M = \frac{L}{f_o} \times \frac{D}{f_e} \Rightarrow 375 = \frac{150}{5} \times \frac{250}{f_e}$

$\Rightarrow f_e = 20\,mm$ Ans.

**EXAMPLE :**

In a compound microscope the focal length of objective lens is 1.2 cm and that of eye piece is 3 cm. When object is placed 1.25 cm in front of objective, final image is formed at infinity. What is the magnification (magnifying power)?

**Solution:** $M = m_o M_e$, By Newton's convention, $m_o = \frac{1.2}{1.25 - 1.2} = 24$

$M_e = D/f_e = 25/3$ thus required magnification: $24 \times \frac{25}{3} = 200$ Ans.

**EXAMPLE :** The focal lengths of the objective and the eye piece of a compound microscope are 2 cm and 3 cm respectively. Distance between objective and eye piece is 15 cm. Final image is formed at ∞. What are $|u_o|$ and $|v_o|$ for the objective lens?

**Solution:** Mechanical tube length, $l_m = v_o + f_e = v_o + 3\,cm = 15\,cm$

$\Rightarrow v_o = 12\,cm$ Ans.

Now $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \Rightarrow \frac{1}{u_o} = \frac{1}{12} - \frac{1}{2} = \frac{-5}{12} \Rightarrow u_o = -2.4\,cm.$

Thus $|u_o| = 2.4\,cm$ and $|v_o| = 12\,cm$.

**EXAMPLE:** In a compound microscope, a magnified virtual image is formed at 25 cm from the eyepiece. Focal length of the objective is 1 cm. If the magnification is 100 and mechanical tube length is 20 cm, what is the focal length of the eyepiece?

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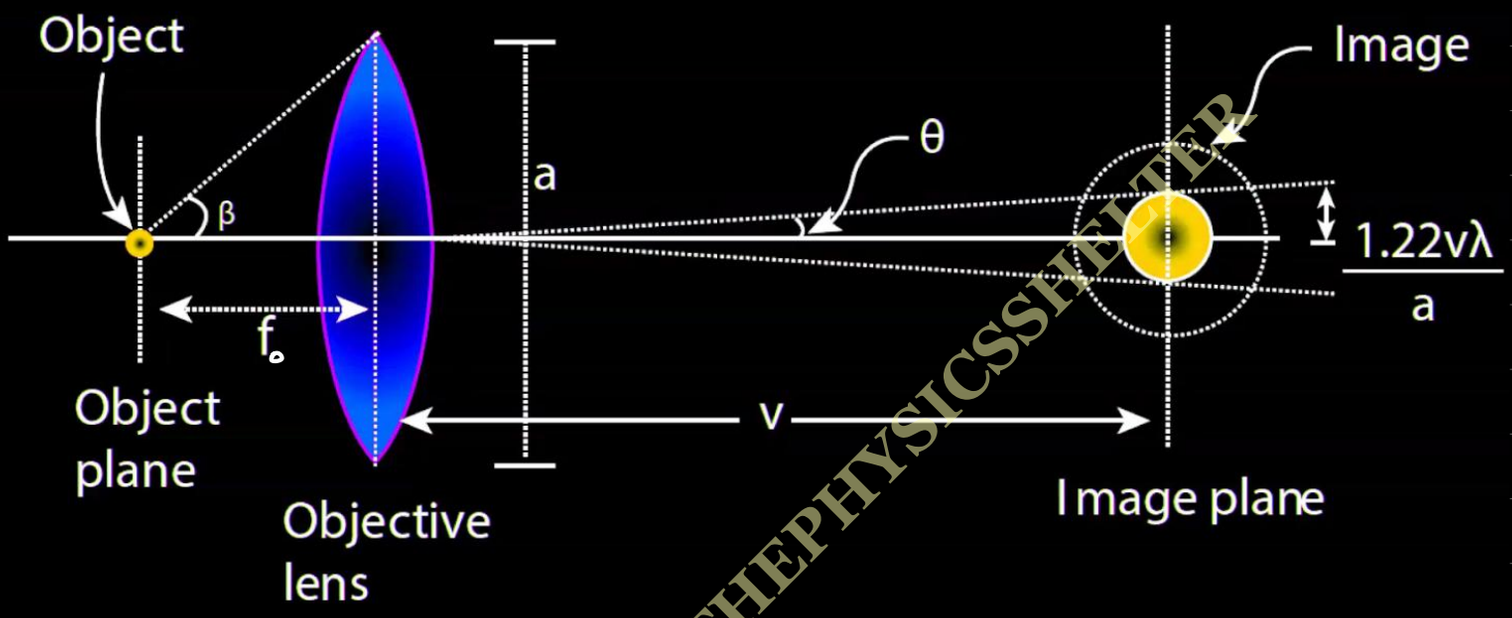
$f_o + x + |u_e| = 20$ | $x = 15.2$ cm
$x/f_o \times D/|u_e| = 100$ | $u_e = 3.8$ cm

**Solution:** $v_o + |u_e| = 20$ cm

$\frac{v_o}{|u_o|} \times \frac{D}{|u_e|} = 100$, also $u_o \approx f_o = 1$ cm ← (Assumption) | $f_e = 4.5$ cm

Solving we get, $u_e = 4\text{cm} = \frac{D f_e}{D + f_e} = \frac{25 f_e}{25 + f_e} \Rightarrow f_e \approx 4.8$ cm.

## Resolving Power Of A Microscope:




Recall from Rayleigh's criterion that for two images to be considered barely resolved the centre of image of one point must lie at the periphery of the other. Now due to diffraction effects if $v$ is the intermediate image position, the radius of the image of the point must be $r_i = \frac{1.22\lambda}{a} \times v$. (∵ angular radius is $1.22\lambda/a$). Thus in the image plane the separation of image centres of two barely resolved images will be $r_i = 1.22\lambda v/a$. Accordingly the required minimum separation in the object plane can be obtained by dividing this by the magnification. Thus if $d_{min}$ is min required separation in the object plane:

$$d_{min} = \frac{r_i}{m_o} = \frac{1.22\lambda}{a} \cdot v \times \frac{|u|}{v}$$

i.e $d_{min} = 1.22\lambda \frac{|u|}{a}$. Now for a typical microscope,

$|u| \approx f_o$ thus $\boxed{d_{min} = 1.22\lambda \frac{f_o}{a}}$

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Now refer to the schematic figure of the ray diagram through the objective lens. Here, $\tan\beta = \frac{a}{2f_o} \approx \sin\beta$ thus we can write

$d_{min} = \frac{1.22\lambda}{2\sin\beta}$. To further reduce the $d_{min}$, the objective lens is immersed in a bath of oil and then,

$$\boxed{d_{min} = \frac{1.22\lambda}{2\mu\sin\beta}}$$ ← Oil immersion objective (Limit Of Resolution)

where $\mu$ is the refractive index of the oil.

The quantity $\mu\sin\beta$ is called numerical aperture i.e:

$$\boxed{N.A. = \mu\sin\beta}$$ thus:

$$\boxed{d_{min} = \frac{1.22\lambda}{2\,N.A.}}$$ Now resolving power is simply $\boxed{R.P. = \frac{1}{d_{min}}}$

**Example:** The numerical aperture of a microscope is 0.12. What is the limit of resolution if viewed in light of wavelength 6000 Å?

**Solution:** $d_{min} = \frac{1.22\lambda}{2NA} = \frac{1.22 \times 6000\,Å}{2 \times 0.12} = 3.05\,\mu m \approx 3\,\mu m$

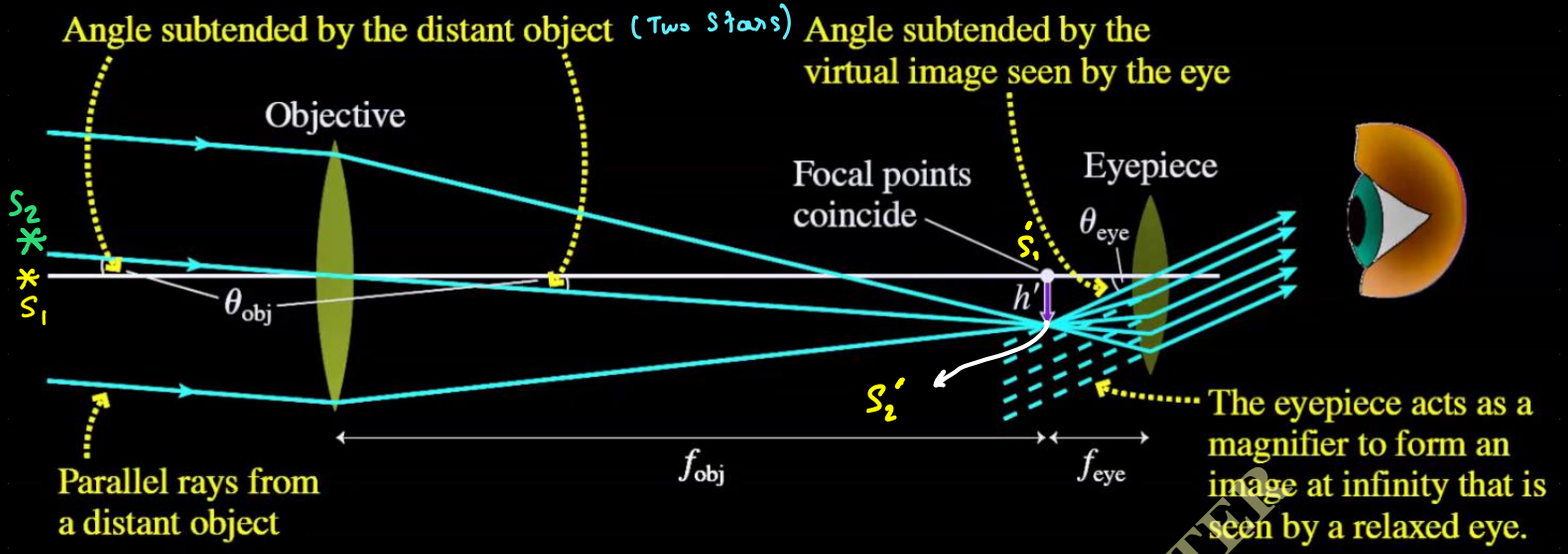
**Example:** An oil immersion objective uses an oil of refractive index $\sqrt{2}$. The illuminating wavelength is 4850 Å and the aperture subtends a 90° angle at the focus. Find the limit of resolution and resolving powers.

**Solution:** $2\beta = 90° \Rightarrow \beta = 45°$. Now $d_{min} = \frac{1.22\lambda}{2\mu\sin\beta} = \frac{1.22 \times 4850\,Å}{2 \times \sqrt{2} \times 1/\sqrt{2}}$

$\Rightarrow d_{min} \approx 3 \times 10^{-7}\,m$ Ans. Thus, Resolving Power: $\frac{1}{d_{min}} = 3.33 \times 10^{6}\,m^{-1}$ Ans.

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# Astronomical Telescope: (Normal Adjustment): (Keplerian Telescope)




The ray diagram above schematically shows the working of a typical refracting astronomical telescope. Let $\theta$ be the angular distance between two distant stars. Their images are formed in the focal plane as shown. The linear distance between their images is $h' = f_o \times \theta$, Now in the final image from the eyepiece, the image of each star appears along the line joining the optical centre of eyepiece to the intermediate images. Thus final image angle: $\theta' = h'/f_e$. Thus angular magnification,

$$M = -\frac{\theta'}{\theta} = \frac{-h'}{f_e} \cdot \frac{1}{\theta} = -\frac{f_o\,\theta}{f_e\,\theta} = -\frac{f_o}{f_e}$$

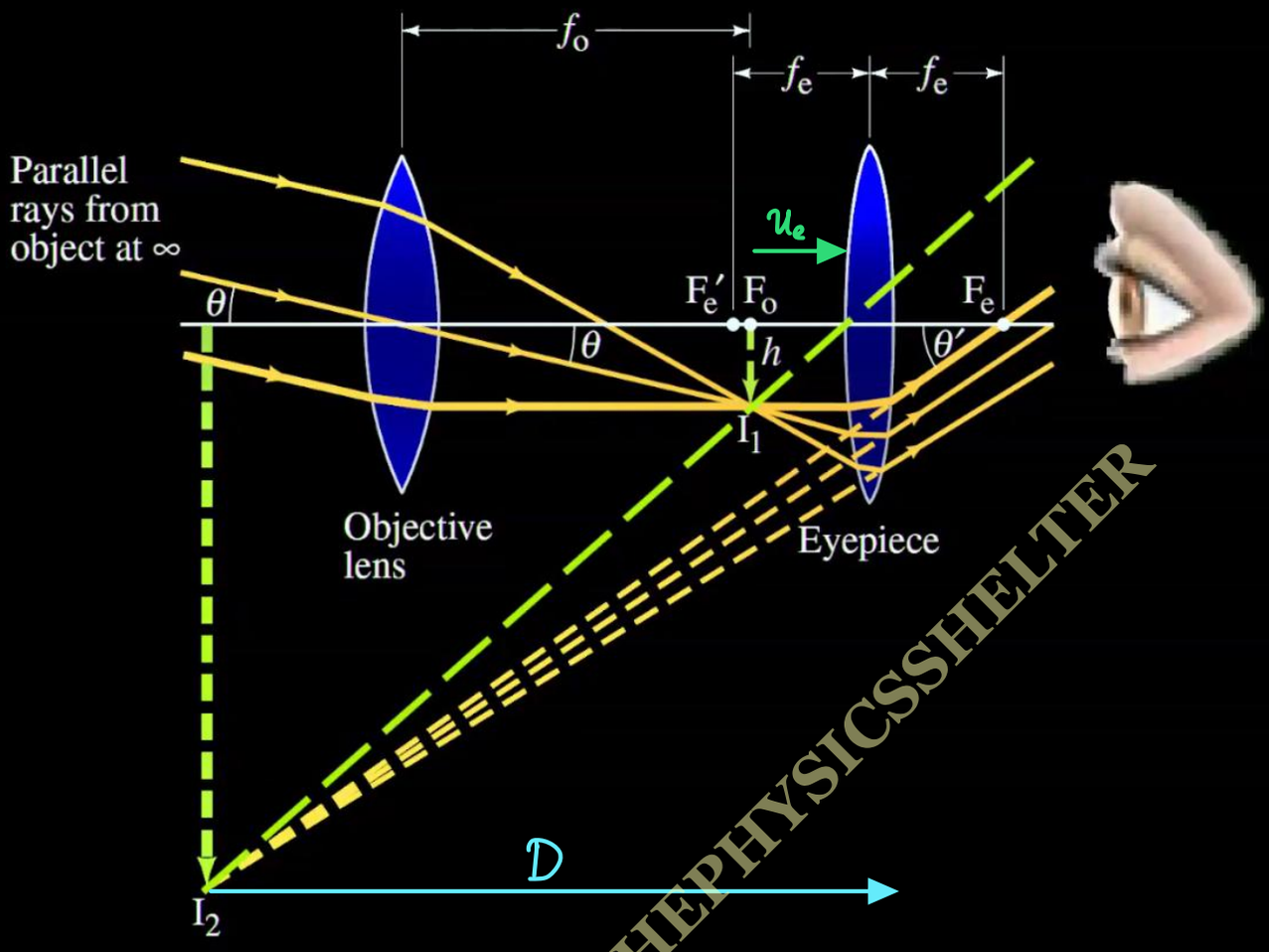
thus $\boxed{M = -\frac{f_o}{f_e}}$ ← Angular magnification in Normal Adjustment. (− sign to signify inverted image).

(Many times −ve sign is dropped in problems, which is fine after all stars don't have to post their smiling pictures on social media!! ☺)

→ Tube length in Normal adjustment is $\boxed{L = f_o + f_e}$

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# Astronomical Telescope: Near Point Adjustment:




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The schematic ray diagram for astronomical telescope in near point adjustment is shown in the figure above. The processing by the objective is exactly same as in case of normal adjustment however eyepiece is adjusted so that $|v_e| = 25\,cm$ and thus $|u_e| = Df_e/(D+f_e)$.

Now $\theta' = h'/|u_e| = f_o\theta/|u_e|$ thus angular magnification:

$$M = -\theta'/\theta = -\frac{f_o}{|u_e|} = \frac{-f_o(D+f_e)}{Df_e} = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)$$

thus:

$$\boxed{M = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)}$$ ← Telescope in near point Adjustment

− Sign for inverted image

→ Many Authors drop the − sign which is fine given that we are interested only in the magnitude of angular magnification.

→ Tube length in near point adjustment is $\boxed{L = f_o + |u_e|}$

**RESOLVING POWER OF A TELESCOPE:** The Resolving Power of a telescope is defined as reciprocal of minimum angular separation required between distant objects to be resolved as per Rayleigh's criterion. As discussed earlier, $\theta_{min} = 1.22\lambda/d$ where d is aperture of the objective. According resolving power is given as:

$$R.P. = 1/\theta_{min} = d/1.22\lambda$$

→ $\theta_{min}$ is also called Limit of Resolution.

**EXAMPLE:** Diameter of objective of a telescope is 250 cm. What is its limit of resolution for light of wavelength 600 nm?

Solution:

$$\Delta\theta_{min} = 1.22\lambda/d = 1.22 \times 600\,nm / 250\,cm = 2.92 \times 10^{-7}\,rad$$

$\approx 3 \times 10^{-7}$ rad Ans.

**EXAMPLE:** An astronomical telescope has angular magnification 5 for distant objects. Tube length is 36 cm (Normal adjustment). What are the focal lengths of two lenses?

Solution: $f_o/f_e = 5$ and $f_o + f_e = 36 \Rightarrow f_o = 30$ cm & $f_e = 6$ cm Ans.

**EXAMPLE:** An astronomical telescope has objective of focal length 200 cm and eyepiece of focal length 4 cm. It is focused on a 10 km far object with near point adjustment. Find the tube length and angular magnification.

Solution: $M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = \frac{200}{4}\left[1 + \frac{4}{25}\right] = 58$ Ans.

Now $M = f_o/|u_e| \Rightarrow |u_e| = f_o/M = 200/58 \approx 3.5$ cm.

$\Rightarrow$ Tube length $= f_o + |u_e| = 203.5$ cm Ans.

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**EXAMPLE:** Find the magnification of a telescope in an arbitrary adjustment where the final image is formed at a distance $l$ from the eyepiece. Focal lengths of objective and eyepiece are $f_o$ & $f_e$.

**Solution:** Here we just need to substitute $D$ by $l$ (see the derivation) thus,

$$\boxed{M = f_o/f_e \left[1 + f_e/l\right]} \quad \text{Ans.}$$

→ Note that for very young people, $l$ can be even less than 25 cm.

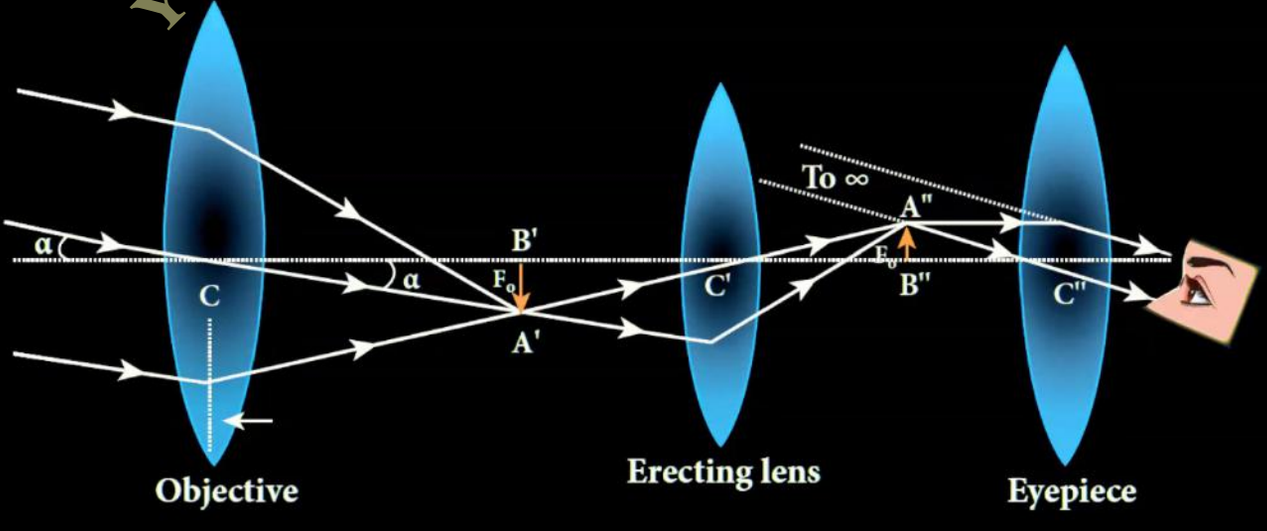
**EXAMPLE:** Find the magnification of a compound microscope in an arbitrary adjustment with final image at a distance $l$.

**Solution:** Here unlike the case of telescope we can't simply replace $D$ by $l$ (in telescopes, $\theta$ was independent of $D$ but with microscope, that is not the case). Now if $h$ is height of the object: $\theta = h/D$. To produce image at a distance $l$, $|u_e| = \left(\frac{lf_e}{l + f_e}\right)$ (∵ there we substituted $D$ by $l$). Thus,

$\theta' = h'/|u_e|$, now angular magnification: $M = \theta'/\theta$ i.e.

$$M = \frac{h'}{h} \cdot \frac{D}{u_e} = \left|\frac{v_o}{u_o}\right| D \left[\frac{1}{f_e} + \frac{1}{l}\right] \quad \text{Ans.}$$

**TERRESTRIAL TELESCOPE:**

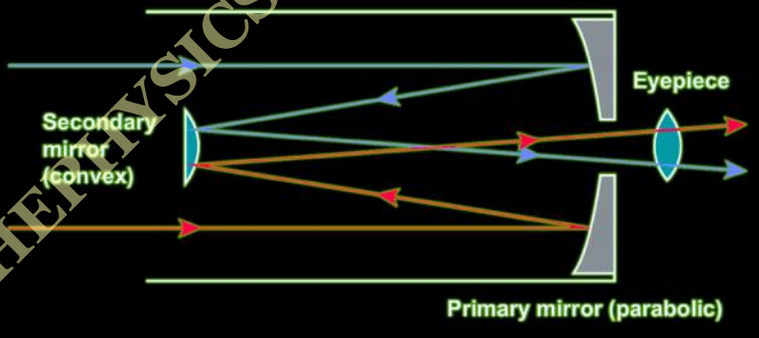



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To observe landscape features on earth such as distant monuments etc, it becomes important to have an erect image. For this purpose, the arrangement similar to astronomical telescope is used but an additional erecting lens is used between the eyepiece and the objective. Here the net magnification and tube length can be found using the logic similar to what we did in Astronomical telescopes. As a special case if the magnification produced by erecting lens is $-1$ then angular magnification is simply $M = f_o/f_e$ (Normal Adjustment) and tube length becomes $l = f_o + f_e + 4f_i$ where $f_i$ is focal length of middle lens.

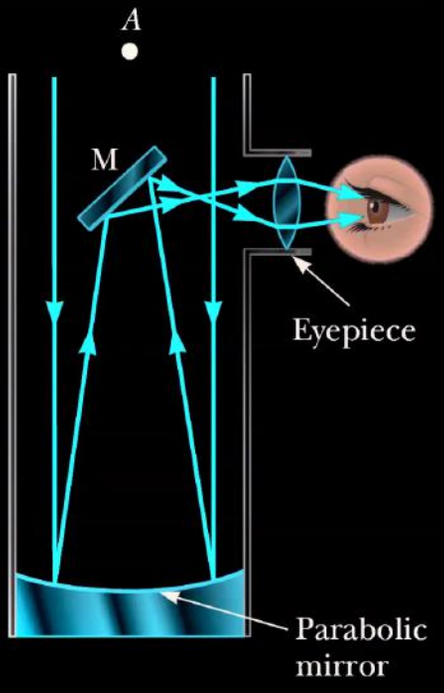
## REFLECTING TELESCOPE (CASSEGRAIN):

In a Cassegrain Reflecting telescope, two mirrors are used as shown in the figure. The magnifications here can be calculated using the procedure similar to that for refracting telescopes. Here if final image is at $\infty$, the magnification is $M = |f_p/f_s|$ where $f_p$ is focal length of primary and $f_s$ is focal length of secondary.




## REFLECTING TELESCOPE: (NEWTONIAN):

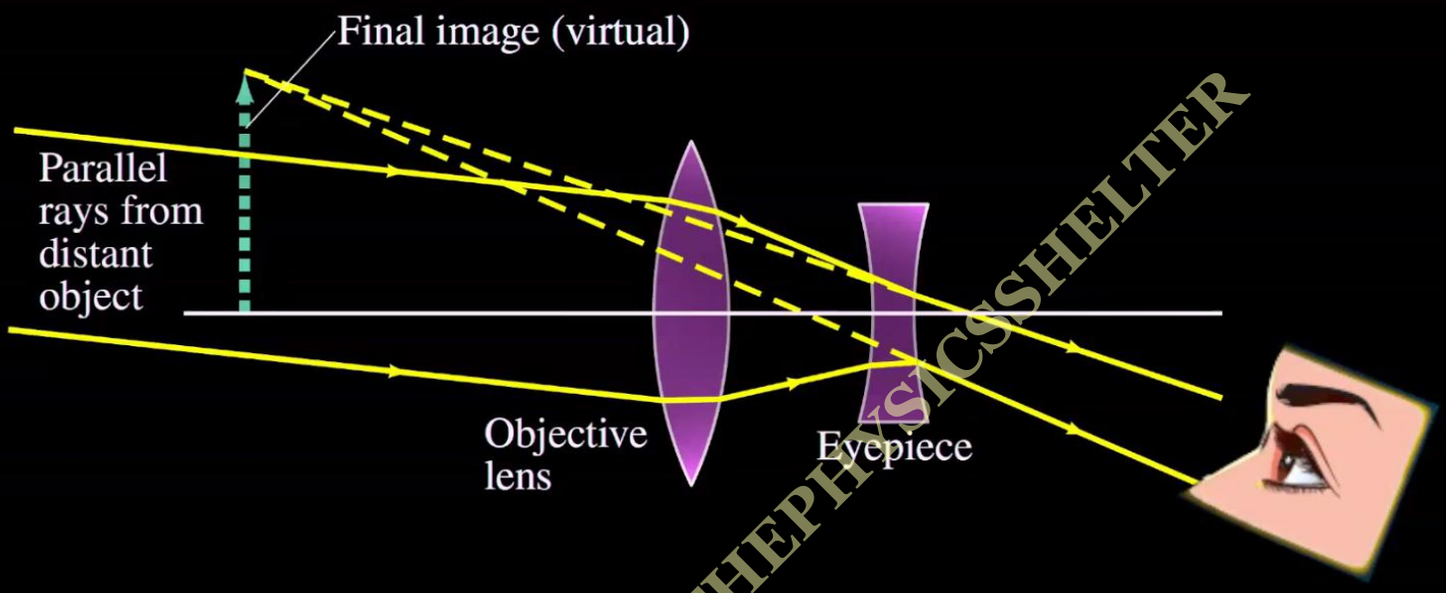
This is the most common telescope used in amateur astronomy clubs. It has a plane mirror oriented at 45° to the principal axis of the parabolic mirror to bend the rays sideways. These deflected rays are then processed using a lens eyepiece. The magnification in this telescope for normal adjustment is similar to that of refracting ones i.e $M = f_o/f_e$ where $f_o$ is the focal length of the parabolic mirror and $f_e$ is focal length of the eye piece.




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→ Reflecting telescopes are often preferred in large observatories because mirrors can be supported nicely at the back using opaque structures, while with huge lenses, due to large weight, supporting them becomes a major engineering challenge (Constraint is to support the lenses without blocking the light paths).

## GALILEAN TERRESTRIAL TELESCOPE:

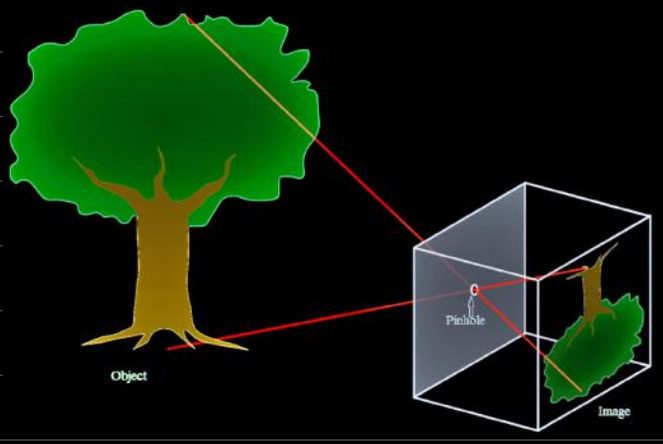



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In this telescope we use one convex and one concave lens. This is the most common arrangement in the amateur binoculars. For normal adjustment, here the tube length will be $L = f_o - |f_e|$ (i.e $f_o + f_e$ ∵ $f_e$ is negative). The angular magnification is $M = |f_o/f_e|$ and with near point adjustment, $M = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$ (Note that $f_e$ is negative). In this case, the tube length is given as $\ell = f_o - |u_e| = f_o + \frac{Df_e}{D+f_e}$.

∵ $f_e$ is negative, Galilean telescope requires smaller tube length for same magnification.

## Pin Hole Camera:




It is a simple box with a hole and a translucent screen to form the image. Since the hole is very small, the image is formed from relatively very small number of rays and thus image is generally very dim. (It is mostly used as a toy or in school scientific demonstration purposes only). The image size can be obtained using similar triangles (knowing the hole to object distance and hole to screen distance). $m = v/u$ where $u$ & $v$ are image and object coordinates wrt pin hole.

**Example: Irodov 5.43:** A Galilean Telescope has 10 fold magnification when focussed at $\infty$ and the tube length is 45 cm. Find (a) Focal lengths of the objective and the ocular (eyepiece) (b) By what distance will ocular have to shift to adjust the telescope to focus at 50 m far object.

**Solution:** $f_o/|f_e| = 10$ and $f_o - |f_e| = 45 \Rightarrow f_o = 50$, $f_e = -5$ Ans.

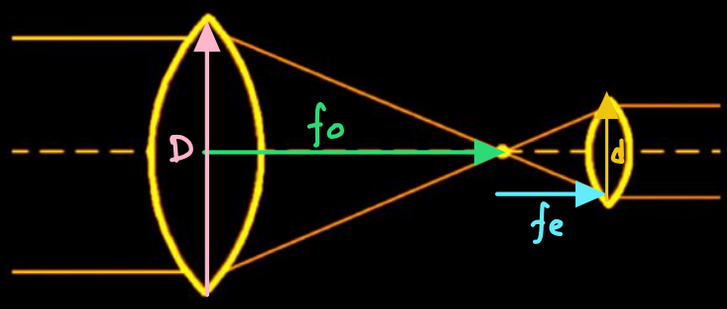
Now if we focus at 50 m away, the distance of intermediate image from focus of objective, $y = \frac{f_o^2}{x} = \frac{50\,cm \times 50\,cm}{(50\,m - 50\,cm)} \approx 0.5\,cm$

When object was at $\infty$, $y = 0$, thus intermediate image is shifted by 0.5 cm towards eyepiece, thus to maintain the adjustment, eyepiece must also be shifted by 0.5 cm Ans.

**Example: Irodov 5.44:** Find the magnification of an Astronomical telescope if the objective diameter is $D$, eyepiece diameter is $d$ and the diameter of eyepiece is just sufficient to collect all the light processed by the objective coming parallel to the principal axis.

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**Solution:** $M = f_o/f_e$. Now consider the adjacent ray diagram. By similar triangles, $f_o/f_e = D/d$. Thus required magnification is simply $D/d$.




→ Notice how the telescope can act as a beam compressor. (Outgoing beam has intensity $D^2/d^2$ times incoming beam intensity, provided that $d$ is just sufficient to accept all light processed by first lens). → $f_o^2/f_e^2$

**EXAMPLE: IRODOV 5.45:** Upon passing through a telescope, intensity of a beam parallel to its principal axis becomes $\eta = 4 \times 10^4$ times. Find the angular dimension of a distant object if its image formed by telescope has an angular dimension $\theta' = 2°$. (eyepiece has sufficient aperture).

**Solution:** $\eta = \left(\frac{D}{d}\right)^2 = \left(\frac{f_o}{f_e}\right)^2 \Rightarrow f_o/f_e = \sqrt{\eta} = M = \theta'/\theta$

Thus $\theta' = \theta/\sqrt{\eta} = 10^{-2}$ degree $= \frac{1}{100}$ degree $=$ 0.6 minute (0.6').

**EXAMPLE: IRODOV 5.46:**

A Keplerian telescope with magnification $\Gamma = 15$ was submerged into water which filled up the inside of the telescope. To make the system work as a telescope again within the former dimensions, the objective was replaced. What has the magnification of the telescope become equal to? The refractive index of the glass of which the ocular is made is equal to $n = 1.50$. (Telescope is still entirely in water)

**Solution:** $M = f_o/f_e = 15 \Rightarrow f_o = 15 f_e \Rightarrow L = 16 f_e$.

Now when eyepiece is surrounded by water its focal length will increase to say $f_e'$. Then:

$$\frac{1}{f_e} = (1.5 - 1)\,K \qquad K: 1/R_1 - 1/R_2$$

$$\frac{1}{f_e'} = \left(\frac{1.5}{4/3} - 1\right) K$$

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$\Rightarrow \frac{f_e'}{f_e} = 4$. Thus eyepiece focal length has increased to $f_e' = 4f_e$

Since tube length is still same $\therefore$ objective focal length : $f_o' = l - f_e'$ i.e $12f_e$. Thus new magnification : $M' = f_o'/f_e' = 12f_e/4f_e = 3$ Ans.

## EXAMPLE: IRODOV 5.48:

The optical powers of the objective and the ocular of a microscope are equal to 100 and 20 D respectively. The microscope magnification is equal to 50. What will the magnification of the microscope be when the distance between the objective and the ocular is increased by 2.0 cm? (Assume that Normal type of adjustment is maintained).

**Solution:** $f_o = \frac{1}{100}\,m = 1\,cm$, $f_e = \frac{1}{20}\,m = 5\,cm$.

Let intermediate image form at $l$ distance from $f_o$ then,

$$\frac{l}{f_o} \times \frac{D}{f_e} = 50 \Rightarrow \frac{l}{1\,cm} \times \frac{25\,cm}{5\,cm} = 50 \Rightarrow l = 10\,cm.$$

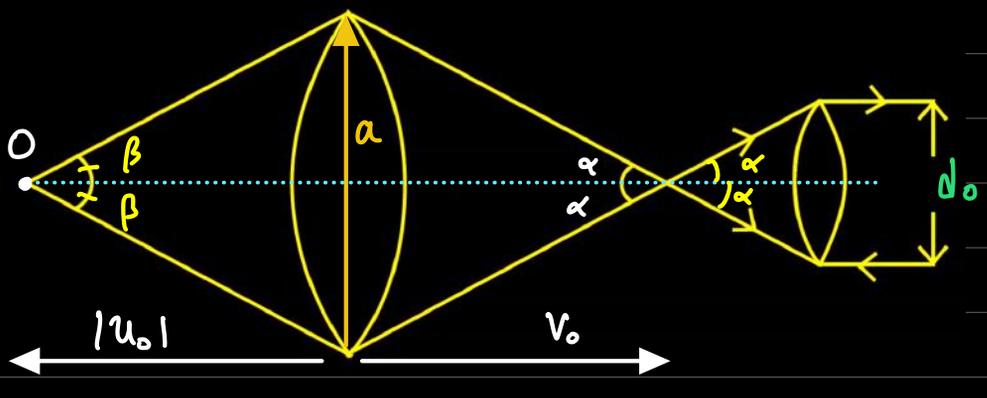
Now $l$ is increased by 2cm thus new value of $l$, $l' = 10+2 = 12\,cm$

Thus new magnification: $\frac{l'}{f_o} \times \frac{D}{f_e} = \frac{12}{1} \times \frac{25}{5} = 60$ Ans.

## EXAMPLE: IRODOV 5.49:

A microscope has a numerical aperture $\sin\beta = 0.12$ where $\beta$ is the half angle subtended by the entrance pupil of the microscope (at the object). Assuming diameter of eye's pupil to be equal to $d_0 = 4\,mm$, estimate the magnification of the microscope at which diameter of beam coming out of microscope is equal to the diameter of eye's pupil.

**Solution:**




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$M = \frac{V_0}{U_0} \times \frac{D}{f_e}$ now from the figure, $d_0 = 2f_e\alpha$ , $a = 2v_0\alpha = 2|u_0|\beta$

thus $\frac{v_0}{|u_0|} = \frac{\beta}{\alpha}$ , thus $M = \frac{\beta}{\alpha} \cdot \frac{D}{f_e}$ , $f_e = \frac{d_0}{2\alpha} \Rightarrow M = \frac{\beta}{\alpha} \times \frac{2\alpha D}{d_0}$

$\Rightarrow M = \frac{2\beta D}{d_0} \approx \frac{2\sin\beta D}{d_0} = \frac{2 \times 0.12 \times 25\,cm}{4\,mm} = 15$ Ans.

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